Al-Balqa Applied University

جامعة البلقاء التطبيقية

Midterm Exam

Course Title	Electrical Circuits (2)	Course Code	ELE 0212
Semester	First Semester	Academic Year	2024-2025
Exam Date	7-12-2024	Day	Saturday
Time	10:00 -10:50		

Student Name _____ Student ID No. _____ Section _____

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Essay	CILO,1,2,3	9(30 %)			
2	Essay	CILO, 3,4	12(40%)			
3	Essay	CILO ,1,2,5	9(30%)			
Total			30(100%)			

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

Note: ***Write all the required equations (without simplification) to solve the following problems, then find the final answer.***

Prob. 1: ***Consider the circuit shown in Fig. 1,***

1. The open circuit voltage between a-b terminals (V_o), when the load impedance is disconnected, is: $(23.41 - j29.2)$ or $(37.48 \angle -51.34)$ V

2. The short circuit current (I_{sc}), when a-b terminals are shorted is: $(4.8 - 9.6j)$ or $(10.733 \angle -63.4)$ A

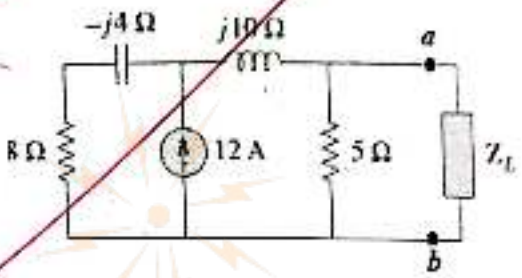


Fig.1

3. If the load impedance $Z_L = 8 + j6\ \Omega$, The load current is:

$$I = \frac{V_L}{Z + Z_L} = \frac{37.48 \angle -51.34}{(8 + j6) + (3.41 + 0.73j)} = 2.82 \angle -81.8\ \text{A}$$

Prob. 2: ***Consider the circuit shown in Fig. 2 and using mesh analysis method,***

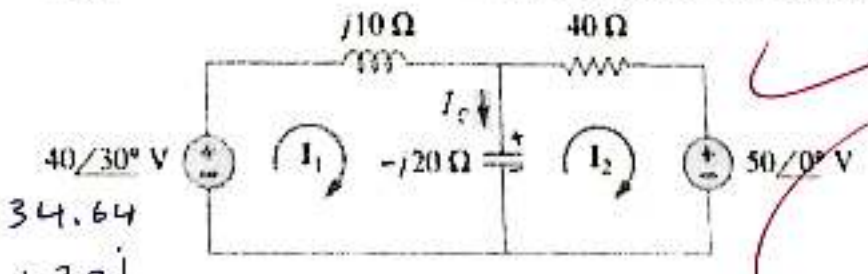
1. Write the two mesh equations, do not simplify:

mesh(1): $-(40 \angle 30) + j10(I_1) + -20j\,(I_1 - I_2) = 0$; $40(I_2) + 50 \angle 0 - 20j(I_2 - I_1)$

2. The current I_1 is: $(-0.428 + j4.678)$ A $(4.69 \angle 95)$ A

3. The current I_2 is: $(0.785 + j\,0.607)$ A

4. The capacitor current, I_c, is: $I_1 - I_2 = (-1.21 + j4.07)$ A $(4.188 \angle 106)$ A



34.64 + 20i

Fig 2

Prob. 3 : ***Consider the circuit shown in Fig. 3.***

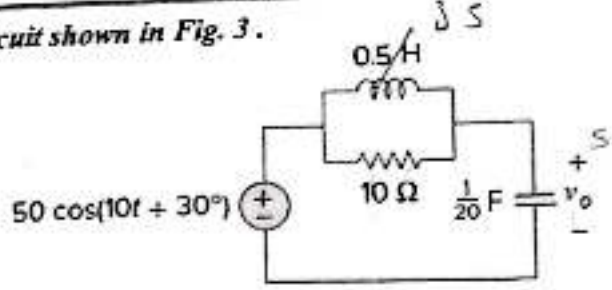


Figure 3

1. The equivalent admittance across the terminals of the source is:

$Z = (2+2i)$ $Y = (0.25 - 0.25i)$

2. The voltage (V0) is: $35.35 \angle -105°$ V

3. The current through 10Ω resistor in phasor domain is:

$(7.9 \angle 48.4°)$ A $7.9\cos(10t + 48.4°)$

4. The total current lags the voltage of the voltage source by :

$17.67 \angle -45°$ degrees

Lags by (45°)

$V_o = \left(\frac{-2j}{2+2i}\right) 50\angle 30°$

$I = \frac{50\angle 30}{2+2i}$

$V_1 = \left(\frac{2+4i}{2+2i}\right) 50\angle 30°$

$V = 79 \angle 48.4$

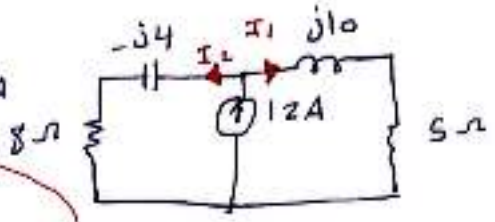
$\frac{I}{1} = \frac{V}{R} =$

$V = IR$

$I = 7.9 \angle 48.4$

نبدأ

1) Voff

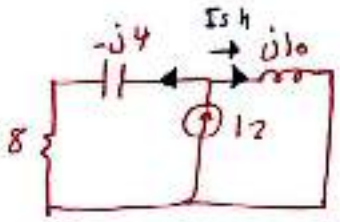


$I_1 = \left(\frac{8-j4}{13+j6}\right) 12 = (4.68 - 5.85j)A$

$V_o = I_1 \times 5\Omega = 23.41 - j29.2\ V$

or $37.48 \angle -51.3\ V$

2) I_{sc}





تعالزي مع الشورت

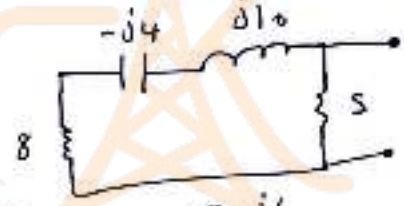
$I_{sc} = \left(\frac{8-j4}{8+j6}\right) 12 = (4.8 - j9.6)A$

or $(10.73 \angle -63)A$

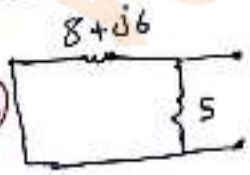
3) I Load : ؟؟

$I_{Load} = \frac{V_{Th}}{Z_{Th} + Z_L}$

Z_{Th}



$8 + j6 \,||\, 5$



$Z_{Th} = (3.41 + j0.73)$

$I_L = \frac{37.48 \angle -51.3}{(8+j6) + (3.41 + j0.73)}$

$I_L = (0.40 - j\,2.799)A$

$= (2.82 \angle -81.8)A$

3

1) write the two mesh equations

$(j10)I_1 + -20j(I_1 - I_2) = 40\angle 30$ mesh 1

$40 I_2 + 50\angle 0 + -j70(I_2 - I_1) = 0$ mesh 2

2) I_1, I_2, I_c

$(-j10 I_1 + j20 I_2 = 40\angle 30) * 2$

$+j20 I_1 + I_2(40 - j20) = -(50\angle 0)$

$-j20 I_1 + j40 I_2 = 80\angle 30$

$+j20 I_1 + I_2(40 - j20) = -(50\angle 0)$

$(40 + j20) I_2 = 44.4\angle 64.2$

$I_2 = \frac{44.4\angle 64.2}{40 + j20} = (0.785 + j0.607)\,A$

$= \boxed{(0.992\angle 37.69)\,A}$ I②

$-j10 I_1 + j70 I_2 = 40\angle 30$

$I_1 = \frac{40\angle 30 - j20(0.992\angle 37.69)}{-j10}$

$= -0.428 + j4.67$

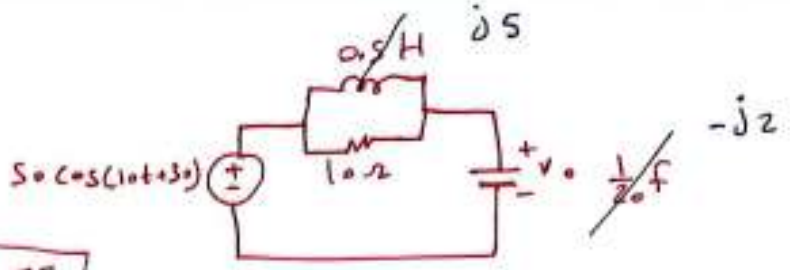
$= \boxed{4.69\angle 95\ A}$ I①

$I_c = I_1 - I_2$

$(4.69\angle 95) - (0.992\angle 37.69) = -1.21 + j4.07$

$= \boxed{4.28\angle 106\ A}$

3)



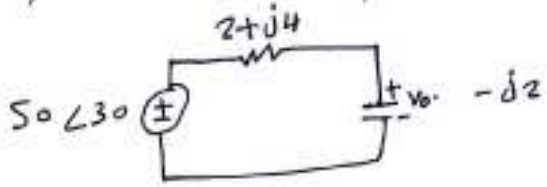
1) $j5 \parallel 10\Omega = (2+j4)\Omega$

$(2+j4) + -j2 = (2+j2)\Omega$

$Y = (2+j2)^{-1} = \boxed{0.25 - j0.25\ \mho}$

$\boxed{0.353\angle -45\ \mho}$

2) V_o



$V_o = \left(\frac{-j2}{2+j2}\right) 50\angle 30$

$= \boxed{35.35\angle -105}\ V$

3) $I_{10\Omega}$

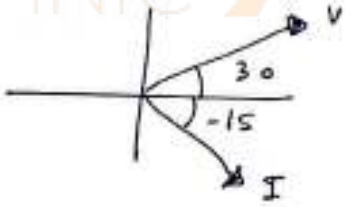
$I_{Total} = \frac{V_{Total}}{Z} = \frac{50\angle 30}{2+2i} = (17.67\angle -15)\ A$

$I_{10\Omega} = \left(\frac{j5}{10+j5}\right) 17.67\angle -15 = 7.9\angle 48.4\ A$

$= \boxed{7.9\cos(10t+48.4)\ A}$

$I_{Total} = 17.67\angle -15$

$V_{Total} = 50\angle 30$



(I) Lags (V) by $\boxed{45^\circ}$